

Algebra II with Trigonometry

Chapter 5.1

Mental Math (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401267_alg-2trig-h-chp-5-1-note-docx.pdf](#)

Degrees to Radians What is the exact radian measure of a right angle?

Radians to Degrees How many degrees are in an angle that measures $\frac{\pi}{3}$ radians?

Coterminal Angles What is the smallest positive angle that is coterminal with 400° ?

Arc Length What is the length of an arc on a circle with a radius of 10 meters that is intercepted by a central angle of 2.5 radians?

Measuring in Radians If an arc of 24 inches subtends a central angle on a circle with a radius of 8 inches, what is the measure of that central angle in radians?

Sector Area Fraction What exact fraction of a circle's total area is represented by a sector with a central angle of π radians?

Area of a Circular Sector Using the formula $A = \frac{1}{2}r^2\theta$, what is the area of a sector if the circle has a radius of 2 cm and the central angle is 3 radians?

Revolutions to Radians If a car wheel completes exactly 5 full revolutions, how many total radians has it rotated?

Angular to Linear Speed A gear with a radius of 4 inches rotates with an angular speed of 3 radians per second. What is the linear speed (in inches per second) of a point on its outer edge?

Coterminal Radians What is the smallest positive angle (in radians) that is coterminal with $\frac{13\pi}{6}$ radians?